JARED CALDO

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EDUCATION

King's College Doha, *Doha, Qatar* 2026–2027

- Pearson Edexcel International A Levels: Mathematics (Predicted A), Physics (Predicted A)
- BTEC Engineering Level 3: Predicted Distinction
- Self-directed study: MIT OpenCourseWare 2.12 Introduction to Robotics

PROJECTS

Mine Sweeper Robot | *ESP32, C++, Autonomous Robotics* Aug–Oct 2026

- Built an autonomous robot on an ESP32 (NodeMCU-32S) that searches a minefield with an inductive proximity sensor, using a search algorithm optimized for shortest clearing time.
- Diagnosed stall-current spikes, voltage sag, and battery internal resistance, and resolved them by moving from AA cells to 18650 Li-ion.
- Documented five design iterations, tracing each failure to its root cause and applying hardware and software fixes.

MicroNAS: Personal Cloud Storage Device | *ESP32, Fusion 360* Aug 2026

- Designed a compact MicroSD-based cloud storage device as a low-cost alternative to HDD-based NAS units; submitted to CUTC Transform Hackathon 2026, placed 6th in the Hardware Track.
- Soldered the circuit, wrote ESP32 firmware for MicroSD access and web integration, and modelled the 3D-printed enclosure.

Automated Temperature Monitoring Box (Commissioned) | *Arduino Uno, C++* Sep–Oct 2026

- Designed, programmed, and assembled a system that detects room entry (PIR), sweeps a DHT11 sensor with a servo, averages three readings, and shows a heating/cooling verdict on an LCD.
- Designed the wiring with the client, wrote the C++ firmware, and built a custom 3D-printed enclosure in Fusion 360.

In progress: OctoCam (digital camera from scratch); automated mini drill feed machine (school capstone project)

EXPERIENCE

Research Manager, *Big Appletunities (Remote)* Aug 2026–Present

- Review opportunities submitted by a team of researchers for accuracy, relevance, and clarity; give feedback and coordinate the research workflow.

Opportunity Researcher, *Big Appletunities (Remote)* Feb–Aug 2026

- Researched and published 60+ scholarships, competitions, and internships for an audience of ~38,000 Instagram followers.

Assistant Basketball Coach, *RTF Basketball, Doha* Jul–Oct 2026

- Co-led weekly training sessions for U10 athletes, focusing on fundamentals skills (ball handling, shooting form, passing, and defense) over 4 months.
- Provided individualized feedback and encouragement, fostering a positive, growth-oriented environment for young athletes.

Program Participant, *GE Aerospace Jetsetters, Doha* Aug 2026

- Participated in on-site job shadowing and work immersion activities at GE Aerospace Qatar, gaining firsthand exposure to aerospace operations and industry practices.
- Engaged with aviation professionals and fellow participants to explore business operations, engineering and safety practices, and career pathways within the aerospace industry.

Team Lead, *Coat Hanger Manufacturing (School Project)* May 2026

- Led a team of three to design and produce two coat hangers in 3 hours using custom 3D printed jigs, applying metal cutting, drilling, lathe work, safety practices, and quality checks.

SKILLS

Programming: C++ (Arduino, ESP32), Python
Hardware: Circuit design, soldering, sensors, servos, battery and power systems
CAD & Fabrication: Fusion 360, 3D printing, mini lathe, hand drill, pillar drill
Other: Iterative design, technical documentation, team management